Data and Applications

Phase 2

Team : Telomere - 37

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EER Model

We have modelled Team 41's requirements in EER_team41_wife_carrying_v2.drawio. The main entities and relationships are represented, but some requirements are incomplete or inconsistent. We have stated the corrections made in the diagram and the limitations that remain. References such as R5, C4 and U-2 refer to Team 41's Phase 1 document, 41.pdf.

Critical Evaluation of the Assigned Document

Completeness

Registration is not fully defined. C4 and I-1 use COMPETITOR_REGISTRATION, while C6, C7 and C15 use REGISTRATION. Their keys, attributes and relationships are not given together anywhere. Bib number, role and weight appear on COMPETITOR, but there is no complete representation of registration status, fee paid, cancellation or refund. The present diagram keeps the declared competitor attributes and leaves this gap unresolved.

Team participation and results are missing. The Team category has three carriers and one carried person, but PARTICIPATES_IN connects only PAIR to RACE_HEAT, and every RACE_RESULT belongs to a PAIR. Therefore the model cannot record a team entering a heat or receiving a result.

Team membership has no edition context. R5 allows competitors to belong to teams over their careers. It also requires three carriers and one carried person at a particular championship. COMPETES_FOR does not include that championship, and C10 uses an undefined TEAM_MEMBER structure. So a team can have members, but the model cannot say which four formed its roster in 2026.

The Sprint Final is not distinguished from qualifying heats. The document describes qualification for a final, and U-1 changes the qualifiers after a timing correction. RACE_HEAT has no round attribute, and no qualification rule is specified. A stored list of qualifiers is not compulsory if it can be derived, but the round and derivation rule must be defined.

Contact and operational records are incomplete. The press form includes journalist details, arrival time, requests and group members, but ACCREDITS records only accreditation_year and drone_permission. PERSON is also described as covering organiser staff and press contacts without the corresponding links. C8, I-2, U-2 and D-1 introduce inspection, pending decision, backpack and audit records without full

definitions. SPONSORSHIP and PRESS_ACCREDITATION may implement the existing SPONSORS and ACCREDITS relationships; their names alone do not require new conceptual entities.

Some catalogue entries are omitted. TEAM_PARTNERSHIP is mentioned in Section 2.1.6, but its referenced Section 2.2.9 does not exist. We have included it using stated assumptions. PROTEST_DECISION appears in the catalogue but is left out of the summary of weak entities. Attribute types and optionality are also incomplete, although domains such as certification_level and sponsorship tier are specified in the prose.

Some attributes are used but never declared. D-1 refers to CHAMPIONSHIP.first_hosted_venue_id, U-2 sets PAIR.status = REPLACED, and C3 relies on backpack_weight_kg. None of these appears in the entity catalogue or in Table 2.1 or Table 2.4 of the phase 1 document. We did not add them to the diagram because the attributes do not have their type, owner or participation defined.

One relationship has two names. R5 names the competitor–team relationship COMPETES_FOR, but C1 enforces the age rule "on insertion into COMPETES_IN". No COMPETES_IN exists. We assumed C1 is meant to apply to registration of a competitor, which is the registration gap described above.

Correctness

Finishing time cannot identify a result. Table 2.2 calls finish_time_ms the partial key of RACE_RESULT, but the full key excludes it. Two entries can have the same time, a time can be corrected, and a non-finisher may have no time. A disqualified entry may still have a recorded time. We therefore show finishing time as an ordinary attribute. Result identity comes from the heat and pair, with at most one result for that combination.

PAIR references use two different key conventions. The catalogue identifies PAIR by (carrier_id, carried_id), while the protest attributes and I-2 use pair_id. We have represented references to PAIR as relationships and retained the ordered carrier-carried combination as its intended identity. This does not require inventing a surrogate identifier. A later relational implementation can reference both components of the composite key.

Bib number and role need context. Section 2.1.2 says bib_number is unique within a championship, whereas C4 says it is unique within a championship and category. Neither is global uniqueness. Removing a global uniqueness tag does not solve the fact that COMPETITOR has no edition context. The single role attribute also conflicts with the statement that one competitor can carry in one pair and be carried in another. The pair relationships already identify those roles.

The disjointness rule has the wrong scope. Section 2.1.2 excludes a person from competing and officiating in the same championship. A globally disjoint specialization excludes that person from doing both at any time. Let us take someone who competed in 2015 and officiated in 2025. The stated event rule allows this, but the present hierarchy does not. We have retained the disjoint hierarchy as written and recorded this limitation.

A single weight does not preserve weigh-ins. weight_kg is recorded at weigh-in but stored once on COMPETITOR. Suppose a carried competitor weighs 48 kg in 2026 and 52 kg in 2027. Replacing the value loses the earlier measurement and changes PAIR.weight_deficit_kg. The measurement needs an event or weigh-in context. Similarly, U-2 must not invalidate all historical pairs and medal results just because the competitor changes role for a later entry.

The protest deadline needs a clock time. C5 compares the current time with the maximum finishing time plus 15 minutes. finish_time_ms is an elapsed duration, such as 64,400 milliseconds. It is not the time of day when the last entry finished. An actual finish or completion timestamp is needed. The document must also clarify whether the deadline concerns the heat or the whole relevant race.

Category ranking needs round rules. The proposed rank_in_category sorts results across all heats of a championship and category. In the Sprint, the same pair can have a qualifying result and a final result. Sorting both together can count that pair twice and produce incorrect final standings. The model needs separate qualification and final-ranking rules, including the treatment of ties, DNF and DQ.

Some status and deadline descriptions disagree. U-3 writes DISQUALIFIED, but the declared result domain uses DQ. It also cites C5 for a 60-minute decision deadline, although C5 defines the 15-minute filing window. C7 and C15 need clearer cancellation wording: C7 includes a forfeiture exception, so it is inaccurate to say that it always blocks late cancellation.

Some apparent duplication needs interpretation. VENUE.track_length_m may mean physical track length, while RACE_CATEGORY.distance_m means race distance. These are not necessarily identical, but one venue attribute cannot describe all courses used there. Nationality and represented country are also not automatically identical. The country medal report needs a policy for mixed-country pairs and must avoid counting a same-country pair twice through its two members.

Two constraints refer to attributes their table does not have. C3 places a CHECK on PAIR, but weight is recorded on COMPETITOR and PAIR holds only the derived weight_deficit_kg. C13 uses protest.pair.championship, but a PAIR has no championship, because a pair can compete in many championships. Both rules need the missing registration structure to be stated correctly.

Modelability

PAIR needs one clear representation. R1 describes CARRIES as a recursive relationship, while PAIR is used as an entity that has attributes and participates in other relationships. We have represented PAIR explicitly, with a separate carrier and carried links to COMPETITOR. The ordered combination must be unique, and the two competitors must be distinct. This avoids representing the same pairing twice.

Participation and results have different lifecycles. PARTICIPATES_IN can record an entry assigned to a heat before its result is available. Therefore retaining it alongside RACE_RESULT is not redundant. The required rule is that every result belongs to a pair that participates in the same heat. The present diagram does not express this cross-relationship condition by its cardinalities alone.

JUDGES needs a constraint on combinations. The ternary relationship records which official judges which category in which championship. There are two rules: every assignment must concern an offered championship-category combination, and every offered combination must have at least one official. Minimum participation on each separate entity cannot guarantee both. The present (1,N) on RACE_CATEGORY also requires every catalogue category to have an assignment somewhere, although HAS_CATEGORY allows unused categories.

Some rules require more than ordinary cardinalities. The team composition rule needs edition context and role counts. Jury-only decisions can be expressed using a suitable official subtype or a written restriction. Protest relationships also need consistency rules so that the protest, concerned heat, filing registration and target result refer to the same championship. These rules should be stated explicitly.

Corrections and Assumptions in the Present Diagram

Pair and result identity. PAIR is shown with HAS_CARRIER and HAS_CARRIED as identifying relationships. Each pair has exactly one competitor in each role, while a competitor can occur in zero or many pairs. RACE_RESULT retains its heat and pair owners, and finish_time_ms is not marked as a partial key. The owner-only identification convention is disclosed in Section 3.

References shown as relationships. NATIONALITY, HOME_COUNTRY, FILED_BY, AGAINST_PAIR, CONCERNS_HEAT and DECIDED_BY replace the corresponding reference attributes. HOSTS and OWNS express the championship venue and organiser associations. This is a conceptual representation choice; it is not a claim that composite foreign keys would cause redundancy in a relational schema.

Other additions and notation. TEAM_PARTNERSHIP is included as a recursive relationship. role_type is included on PERSON because it is described in the source prose. heat_number, protest_seq and decision_seq use dashed underlining. bib_number is not marked globally unique; its correct event scope remains unresolved.

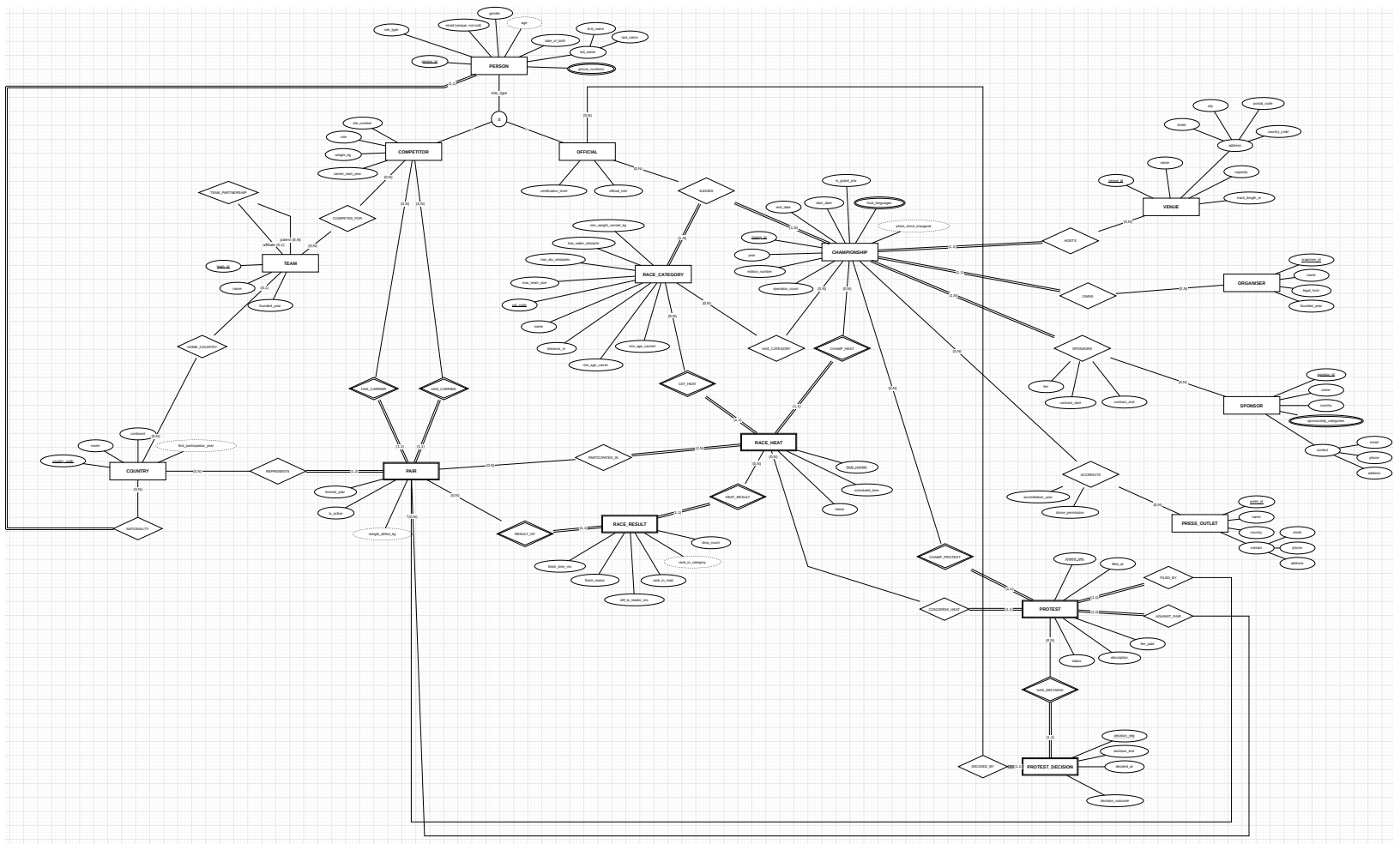
Participation assumptions. NATIONALITY gives each PERSON exactly one COUNTRY. HOME_COUNTRY permits a TEAM to have zero or one home country. Every PROTEST has one filing pair, one targeted pair and one concerned heat. Every PROTEST_DECISION has one deciding official. A team may have zero or one parent team and may be parent to many affiliates. These minima are assumptions where the source does not give a complete rule; a foreign key alone does not establish mandatory participation.

The present diagram retains the source's global disjointness, competitor role and weight attributes, venue track length, pair-only results, and stored heat rank and time difference. These limitations are evaluated here rather than described as completed fixes.

Notational Deviations

These are required but cannot be drawn in the software we used (draw.io), so they are stated here:

→A pair's carrier and carried competitor must be different people, and the ordered combination must be unique.
→Only an official with the role JURY may decide a protest, and the protest, concerned heat, filing pair and targeted pair must belong to the same championship.
→Every result must belong to a pair that participates in the same heat.
→Every (championship, category) in JUDGES must also appear in HAS_CATEGORY.
→A team has three carriers and one carries per championship, which needs the missing edition context.
→Age and weight rules and the protest and cancellation time windows.


PERSON
COMPETITOR
OFFICIAL
JUDGES
TEAM
COUNTRY
NATIONALITY
REPRESENTS
PAIR
TEAM_PARTNERSHIP
COMPETES_FOR
HOME_COUNTRY
TEAM_CARRIES
DUO_CARRIES
RACE_CATEGORY
CHAMPIONSHIP
HOSTS
CHAIRS
ORGANISER
SPONSORS
SPONSOR
VENUE
ACCREDITS
PRESS_OUTLET
RACE_HEAT
PARTICIPATES_IN
HEAT_RESULT
RACE_RESULT
RESULT_OF
CHAMP_PROTEST
CONCERNS_HEAT
PROTEST
FILED_BY
AGAINST_PAIR
JUDGES_VOTE
DECIDES_ON
PROTEST_DECISION
DECIDED_BY

Design Quality Analysis

Redundancy Analysis

RACE_RESULT stores rank_in_heat and diff_to_leader_ms even though they can be calculated from eligible results and the applicable ranking rules. These are repeated derived facts. If they remain stored, every correction affecting the order must update them consistently. Showing a derived attribute in the diagram is not itself redundant storage; this concern applies to the attributes deliberately kept as stored values.

COMPETITOR.role also overlaps with HAS_CARRIER and HAS_CARRIED. Let us take a competitor who carries in one pair and is carried in another. The relationships record both facts, but a single role value cannot agree with both. Therefore the role should belong to the particular entry or membership.

PARTICIPATES_IN and RACE_RESULT are not necessarily duplicates because a scheduled participation can exist without a result. Likewise, REPRESENTS is not derivable from nationality unless the model explicitly defines representation that way. accreditation_year is redundant only if it means the year of the linked championship, rather than the year when accreditation was approved.

Update Anomaly

Suppose one pair's time is corrected from 36.10 seconds to 35.80 seconds and it moves ahead of another pair. Updating finish_time_ms alone leaves the stored ranks and time differences incorrect. We have to update several RACE_RESULT records for one timing correction. This is the clearest update anomaly in the present model.

Weight creates a separate history problem. Updating COMPETITOR.weight_kg from 48 to 52 changes the derived deficit for every pair using that competitor, even if an earlier race required ballast. The single value cannot preserve multiple weigh-ins. Role and roster changes must also be limited to the affected event entry so that valid historical results remain unchanged.

Insertion Anomaly

CHAMPIONSHIP has mandatory participation in SPONSORS. Therefore we cannot record a new championship placeholder until at least one sponsor association exists. However, D-1 in the source allows an empty placeholder without sponsorships. If we are entitled to create that placeholder first, the diagram unnecessarily makes its insertion depend on sponsorship information. This is a participation restriction, rather than a demonstrated normalization failure.

Similarly, RACE_HEAT requires at least two participating pairs. This prevents us from recording a scheduled heat that is still accepting entries, or retaining a cancelled heat with fewer than two entries. The minimum should apply when a heat is confirmed or started, rather than throughout its existence. Mandatory unique email is another limitation when a historical athlete is known but their email is unavailable.

Deletion Anomaly

The diagram does not state a deletion policy, but its weak entities imply one. PAIR depends on COMPETITOR through HAS_CARRIER and HAS_CARRIED, and RACE_RESULT depends on PAIR and RACE_HEAT. In the standard E&N mapping, deleting an owner removes its dependents. So deleting a competitor (for example, to honour a data-removal request) would also remove every pair that person was

part of, and through those pairs their results. The partners would lose valid results that have nothing to do with the deleted person. Deletion should therefore be restricted while historical references exist, with withdrawal or archiving handled separately.

Deleting the last SPONSORS association creates another issue: the championship would violate its mandatory sponsorship participation. This should not cause the championship or its race history to be deleted. The intended lifecycle and deletion policy must be stated. With restricted deletion and independent history records, the model can avoid that loss; the present diagram alone does not guarantee it.

Use of Generative AI

Generative AI was Used to verify consistencies with team 41's submission, and verify the draw.io image's consistency and identify any redundant/zero-length vector objects in the EER diagram. It was also used to identify the notational differences exhaustively.

It was NOT used to analyze any of the contents from 41.pdf and make any assessments. It was NOT used to generate the EER diagram.